

1

Let $\pi = (t_0, \dots, t_n)$ be a partition of $[t - \tau, t]$ with $\Delta t_k := t_k - t_{k-1}$ and log-returns $\Delta \ln S_{t_k} := \ln S_{t_k} - \ln S_{t_{k-1}}$. Assume

$$d \ln S_u = b_u dW_u, \quad \text{and on each } [t_{k-1}, t_k] \text{ we have } b_u \equiv b_k \text{ (piecewise constant).}$$

Define

$$\sigma_t^\pi = \sqrt{\frac{1}{\tau} \sum_{k=1}^n (\Delta \ln S_{t_k})^2}, \quad \text{RV}_t^\pi := \frac{1}{\tau} \sum_{k=1}^n (\Delta \ln S_{t_k})^2.$$

By Itô isometry on $[t_{k-1}, t_k]$,

$$\Delta \ln S_{t_k} = \int_{t_{k-1}}^{t_k} b_k dW_u \sim \mathcal{N}(0, b_k^2 \Delta t_k), \quad \Rightarrow \quad (\Delta \ln S_{t_k})^2 \stackrel{d}{=} b_k^2 \Delta t_k Z_k^2, \quad Z_k \stackrel{i.i.d}{\sim} \mathcal{N}(0, 1).$$

Hence

$$\text{RV}_t^\pi = \frac{1}{\tau} \sum_{k=1}^n b_k^2 \Delta t_k Z_k^2.$$

Mean and variance of the realized variance RV_t^π . Using $\mathbb{E}[Z_k^2] = 1$ and $(Z_k^2) = 2$,

$$\mathbb{E}[\text{RV}_t^\pi] = \frac{1}{\tau} \sum_{k=1}^n b_k^2 \Delta t_k = \frac{1}{\tau} \int_{t-\tau}^t b_u^2 du =: \text{IV}_t$$

$$(\text{RV}_t^\pi) = \frac{1}{\tau^2} \sum_{k=1}^n b_k^4 \Delta t_k^2 (Z_k^2) = \frac{2}{\tau^2} \sum_{k=1}^n b_k^4 \Delta t_k^2$$

(the cross-terms vanish by independence).

Bias and variance of $\sigma_t^\pi = \sqrt{\text{RV}_t^\pi}$ via the delta method. Let $g(y) = \sqrt{y}$. A 2nd-order expansion at $\mu := \text{IV}_t$ gives

$$\mathbb{E}[g(\text{RV}_t^\pi)] \approx g(\mu) + \frac{1}{2} g''(\mu) (\text{RV}_t^\pi), \quad (g(\text{RV}_t^\pi)) \approx (g'(\mu))^2 (\text{RV}_t^\pi),$$

with $g'(\mu) = \frac{1}{2\sqrt{\mu}}$, $g''(\mu) = -\frac{1}{4\mu^{3/2}}$. Therefore

$$(\sigma_t^\pi) := \mathbb{E}[\sigma_t^\pi] - \sqrt{\text{IV}_t} \approx \frac{1}{2} g''(\text{IV}_t) (\text{RV}_t^\pi) = -\frac{1}{8} \frac{(\text{RV}_t^\pi)}{\text{IV}_t^{3/2}} = -\frac{1}{4\tau^2} \frac{\sum_{k=1}^n b_k^4 \Delta t_k^2}{\text{IV}_t^{3/2}} < 0$$

$$(\sigma_t^\pi) \approx (g'(\text{IV}_t))^2 (\text{RV}_t^\pi) = \frac{1}{4\text{IV}_t} (\text{RV}_t^\pi) = \frac{1}{2\tau^2 \text{IV}_t} \sum_{k=1}^n b_k^4 \Delta t_k^2$$

(Optional exact check: constant b and equal spacing). If $b_u \equiv b$ and $\Delta t_k = \tau/n$, then

$$\sigma_t^\pi = b \sqrt{\frac{1}{n} \sum_{k=1}^n Z_k^2} = b \sqrt{\frac{\chi_n^2}{n}},$$

so

$$\mathbb{E}[\sigma_t^\pi] = b \frac{\sqrt{2} \Gamma(\frac{n+1}{2})}{\sqrt{n} \Gamma(\frac{n}{2})}, \quad (\sigma_t^\pi) = b^2 \left(1 - \left[\frac{\sqrt{2} \Gamma(\frac{n+1}{2})}{\sqrt{n} \Gamma(\frac{n}{2})} \right]^2 \right),$$

confirming negative bias and agreement with the delta-method as $n \rightarrow \infty$.

How bias/var change Since $\sum_k \Delta t_k = \tau$ and $\sum_k \Delta t_k^2 \leq \tau \|\pi\|$:

- If $\|\pi\| \downarrow 0$ with τ fixed, then $\sum_k \Delta t_k^2 = O(\|\pi\|)$ so

$$|(\sigma_t^\pi)|, (\sigma_t^\pi) = O\left(\frac{\|\pi\|}{\tau^2}\right) \rightarrow 0.$$

- If $\tau \uparrow$ while $\|\pi\| = h$ is held fixed (typical steps $\approx h$), then $\sum_k \Delta t_k^2 \asymp \tau h$ and IV_t is an average (order 1), so

$$(\text{RV}_t^\pi) = O\left(\frac{h}{\tau}\right), \quad |(\sigma_t^\pi)|, (\sigma_t^\pi) = O\left(\frac{h}{\tau}\right).$$

2

Model and two pricings. Under $\mathbb{Q}$,

$$dS_t = rS_t dt + \sigma(t)S_t dW_t, \quad \text{with deterministic } \sigma(\cdot).$$

Over $[t, T]$ the log-price is

$$\ln S_T = \ln S_t + \left(r - \frac{1}{2}\sigma_{\text{eff}}^2(t)\right)(T-t) + \sigma_{\text{eff}}(t)\sqrt{T-t}Z,$$

where

$$\sigma_{\text{eff}}^2(t) := \frac{1}{T-t} \int_t^T \sigma^2(u) du, \quad Z \sim \mathcal{N}(0, 1).$$

Thus any European option admits the Black–Scholes formula with volatility $\sigma_{\text{eff}}(t)$ (“time-varying BS”). The comparator is standard BSM with the constant volatility

$$\sigma := \sqrt{\frac{1}{T} \int_0^T \sigma^2(u) du}$$

(the initial-time average).

Delta at the initial time $t = 0$. Time-varying model: $\Delta^{\text{tv}}(0) = N(d_1(0; \sigma_{\text{eff}}(0)))$ with

$$\sigma_{\text{eff}}(0) = \sqrt{\frac{1}{T} \int_0^T \sigma^2(u) du} = \sigma.$$

Constant-vol model: $\Delta^{\text{c}}(0) = N(d_1(0; \sigma))$.

$$\Delta^{\text{tv}}(0) = \Delta^{\text{c}}(0)$$

because the effective and constant vols coincide at $t = 0$.

Delta at an arbitrary $t > 0$.

$$\Delta^{\text{tv}}(t) = N\left(\frac{\ln(S_t/K) + (r + \frac{1}{2}\sigma_{\text{eff}}^2(t))(T-t)}{\sigma_{\text{eff}}(t)\sqrt{T-t}}\right), \quad \Delta^c(t) = N\left(\frac{\ln(S_t/K) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}\right).$$

These are equal iff $\sigma_{\text{eff}}(t) = \sigma$ (which holds exactly when $\sigma(\cdot)$ is constant, or more generally when the remaining-time average equals the full-time average).

In general, $\Delta^{\text{tv}}(t) \neq \Delta^c(t)$ for $t > 0$.

Theta difference and explicit calculation. Let $C^{\text{BS}}(S, \tau, \sigma)$ denote the Black–Scholes price with time to maturity $\tau = T - t$.

$$C^{\text{tv}}(t) = C^{\text{BS}}(S_t, \tau, \sigma_{\text{eff}}(t)), \quad C^c(t) = C^{\text{BS}}(S_t, \tau, \sigma).$$

Differentiate $C^{\text{tv}}(t)$ w.r.t. t using the chain rule:

$$\Theta^{\text{tv}}(t) := \frac{d}{dt}C^{\text{tv}}(t) = \underbrace{\frac{\partial C^{\text{BS}}}{\partial t}(S_t, \tau, \sigma_{\text{eff}}(t))}_{\text{standard BS theta with } \sigma_{\text{eff}}(t)} + \underbrace{\text{Vega}^{\text{BS}}(S_t, \tau, \sigma_{\text{eff}}(t)) \frac{d\sigma_{\text{eff}}(t)}{dt}}_{\text{extra term}}.$$

Here $\text{Vega}^{\text{BS}} = \partial C^{\text{BS}} / \partial \sigma$. To make the extra term explicit, write $v(t) := \sigma_{\text{eff}}^2(t) = \frac{A(t)}{B(t)}$ with

$$A(t) := \int_t^T \sigma^2(u) du, \quad B(t) := T - t.$$

Then $A'(t) = -\sigma^2(t)$ and $B'(t) = -1$, so by the quotient rule

$$v'(t) = \frac{A'B - AB'}{B^2} = \frac{-(\sigma^2(t))(T-t) - \left(\int_t^T \sigma^2(u) du\right)(-1)}{(T-t)^2} = \frac{\int_t^T \sigma^2(u) du - (T-t)\sigma^2(t)}{(T-t)^2}.$$

Since $\sigma_{\text{eff}}(t) = \sqrt{v(t)}$,

$$\frac{d\sigma_{\text{eff}}(t)}{dt} = \frac{v'(t)}{2\sqrt{v(t)}} = \frac{\int_t^T \sigma^2(u) du - (T-t)\sigma^2(t)}{2(T-t)^2 \sigma_{\text{eff}}(t)}$$

and therefore

$$\Theta^{\text{tv}}(t) = \Theta^{\text{BS}}(S_t, \tau, \sigma_{\text{eff}}(t)) + \text{Vega}^{\text{BS}}(S_t, \tau, \sigma_{\text{eff}}(t)) \cdot \frac{\int_t^T \sigma^2(u) du - (T-t)\sigma^2(t)}{2(T-t)^2 \sigma_{\text{eff}}(t)}$$

whereas

$$\Theta^c(t) = \Theta^{\text{BS}}(S_t, \tau, \sigma)$$

has no extra term because σ is constant. Thus thetas are generally different; they coincide iff $\frac{d\sigma_{\text{eff}}(t)}{dt} = 0$, e.g. when $\sigma(\cdot)$ is (essentially) constant on $[t, T]$.